

13.3 Arc Length and Curvature

1 Warm Up

Question 1: How can I find the length of a curve from calculus II?

2 Motivation

3 Definitions & Theorems

1. In two dimensions, using parametric equations $x = f(t)$ and $y = g(t)$, we get the length of an arc from $a \leq t \leq b$ to be

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt$$

2. For a space curve, our length of a curve formula is really no different. We now have three dimensions. Suppose our space curve has the vector function $\langle f(t), g(t), h(t) \rangle$, the length of the space curve L is as follows.

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt = \int_a^b |r'(t)| dt.$$

3. We can now define an **arc length function** $s(t)$ which just takes our integral from an initial value a to end point t which is variable.

$$s(t) = \int_a^t |r'(u)| du$$

Differentiating both sides we see that $ds/dt = |r'(t)|$ by the fundamental theorem of calculus. Since we now have s as a function of t , i.e. a correspondence between s and t , we can solve for t and **re-parameterize** our vector function in terms of s .

4. If we want to consider how quickly a curve changes direction, it is natural to look at the rate at which the derivative changes with respect to the arc length at that given time. We call this **curvature** and it is defined as follows.

$$k = \left| \frac{dT}{ds} \right| = \left| \frac{dT/dt}{ds/dt} \right| = \left| \frac{T'(t)}{r'(t)} \right|.$$

5. Using our derivative rules (product rule) there is a very cool way to write curvature. The proof to the following theorem is omitted.

Theorem 1. (10 in book) $k = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}$

6. Our unit tangent vector and arc length re-parameterization can help us out in another way. The entire purpose of our standard basis representation $\hat{i}, \hat{j}, \hat{k}$ is that every vector can be written as a linear combination of these. The only thing special about these though is that they are unit, and mutually orthogonal. They give us a global view of a space curve as if someone is looking at it from afar and can see the origin and everything as if its on a movie screen. What if instead there was a new basis system which was more beneficial from the perspective of the curve. If we think about a plane flying through a space curve, the pilot doesn't care much about i's, j's, and k's but rather their own view point. We therefore define the following vectors. We call them the **TNB-frames**

- **T**: Unit Tangent Vector (Direction, nose of plane)
- **N**: Unit Normal Vector (Turning, tail-fin of plane)
- **B**: Unit Binormal Vector (Twist, wings of plane)

$$T(t) = \frac{r'(t)}{|r'(t)|} \quad N(t) = \frac{T'(t)}{|T'(t)|} \quad B(t) = \frac{T(t) \times N(t)}{|T(t) \times N(t)|} = T(t) \times N(t).$$

4 Examples

Problem 1. Find the arc length of the vector function $\vec{r}(t) = \langle 2t, 3 \sin(2t), 3 \cos(2t) \rangle$ from $0 \leq t \leq 2\pi$.

Problem 2. Find the arc length of the vector function $\vec{r}(t) = \frac{1}{3}t^3\hat{i} + 4t\hat{j} + \sqrt{2}t^2\hat{k}$ from $0 \leq t \leq 2$.

Problem 3. Find the arc length function of the vector function from Problem 2.

Problem 4. Reparameterize the following vector function in terms of arc length.

$$\vec{r}(t) = \langle e^t \cos(t), e^t \sin(t), e^t \rangle$$

Problem 5. Find the curvature of the vector function $\vec{r}(t) = \langle 4t, -t^2, 2t^3 \rangle$.

Problem 6. Find $T(t)$, $N(t)$, and $B(t)$ if $r(t) = \cos(3t)\hat{i} + \sin(3t)\hat{j} + 5\hat{k}$.

5 Scratch Work